Bernoulli Distribution

- Imagine conducting an experiment where you repeat a certain action n times, and each time, it either succeeds or fails.
- We'll call each of these attempts a "trial." If we label X_j as 1 when the jth trial succeeds and 0 when it fails, we're dealing with what's known as Bernoulli trials.
- Now, if these trials follow a specific set of rules, we call them a Bernoulli process. Here's what makes it a Bernoulli process:
    - The trials are independent (meaning the outcome of one doesn't affect the others), each trial has only two possible outcomes (success or failure), and the probability of success remains the same across all trials.
    - For each trial, we can express the probability of success or failure using a function $P(x_j)$. This function is pretty straightforward: if x_j is 1 (meaning success), the probability $P(x_j) = p$ and if x_j is 0 (indicating failure), the probability is $1 - p$

- Now, let's talk about the Bernoulli distribution. This is simply the probability distribution for one trial. It tells us the likelihood of success or failure in a single attempt.
- The mean $E(X_j)$ gives us the average outcome over many trials. For a Bernoulli trial, it's just the probability of success, which is p.
- The variance $V(X_j)$ measures how much the outcomes vary from the mean. For Bernoulli trials, it's calculated using the formula $p(1 - p)$.

Binomial Distribution

- The Binomial distribution comes into play when we're interested in counting the number of successes in a fixed number n of Bernoulli trials.
- Imagine you have n independent Bernoulli trials, each with a probability p of success and $q = 1 - p$ of failure.

- Now, let X be the random variable that represents the total number of successes among these n trials.
- The probability $p(x)$ of observing x successes out of n trials is given by the equation below:

$$p(x) = \begin{cases} \binom{n}{x} p^x q^{n-x}, & x = 0, 1, 2, \ldots, n \\ 0, & \text{otherwise} \end{cases}$$

- It tells us that the probability of observing x successes is $p^x . q^{n-x}$, where q is the probability of failure and p is the probability of success.
- Equation above might seem daunting at first, but it's motivated by a straightforward idea: we're calculating the probability of a specific outcome where we have x successes (denoted by S) followed by $n - x$ failures (denoted by F). This calculation involves combinatorics, hence the term $\binom{n}{x}$, which represents the number of combinations of x successes in n trials, or number of ways we can choose x successes out of n trials.
- $\binom{n}{x} = \frac{n!}{x!(n-x)!}$, or n nCr x on a calculator.



- Since X is the sum of n independent Bernoulli random variables (each with mean p and variance $p(1 - p) = pq$), the mean $E(X)$ is np as the Equation below shows.

$$E(X) = p + p + \ldots + p = np$$

- Similarly, the variance npq as shown in Equation below.

$$V(X) = pq + pq + \ldots + pq + pq = npq$$

Example

- Consider a production process that manufactures computer chips, with an average nonconforming rate of 2%.
- Each day, a random sample of 50 chips is taken from the process. If this sample contains more than two nonconforming chips, the process will be stopped.
- Let's calculate the probability that the process is stopped based on this sampling scheme.
- We can model the sampling process as $n = 50$ Bernoulli trials, where each trial has a success probability $p = 0.02$, representing the probability of selecting a nonconforming chip.
- The total number of nonconforming chips in the sample, denoted by X, follows a binomial distribution. The probability mass function (PMF) of X is given by Equation below, which we've already discussed

$$p(x) = \begin{cases} \binom{n}{x} p^x q^{n-x}, & x = 0, 1, 2, \ldots, n \\ 0, & \text{otherwise} \end{cases}$$

- To find the probability that more than two nonconforming chips are found in a sample, we can use the complement rule: $P(X > 2) = 1 - P(X \leq 2)$.
- Now, we need to calculate $P(X \leq 2)$, which represents the probability of observing at most two nonconforming chips in the sample. We can do this by summing the probabilities of $X = 0$, $X = 1$, $X = 2$.

$$P(X \leq 2) = P(X = 2) + P(X = 1) + P(X = 0)$$

$$= (0.02)^2(0.98)^{48}1225 + (0.02)(0.98)^{49}50 + (0.98)^{50} = 0.92$$

- Therefore the probability that the production process is stopped on any given day, based on the sampling process, is approximately $1 - 0.92 = 0.08$.
- In a random sample of size 50, the mean number of nonconforming chips (successes) is calculated to be 1, and the variance is approximately 0.98:

$$E(X) = np = 50(0.02) = 1$$
$$V(X) = npq = 50(0.02)(0.98) = 0.98$$

Geometric and Negative Binomial Distributions

Geometric Distribution

- This distribution represents the number of Bernoulli trials needed to achieve the first success. For each trial, there are two possible outcomes: success (with probability p) or failure (with probability $q = 1 - p$)
- The probability mass function $p(x)$ for the geometric distribution is given by $q^{x-1}p$ where x represents the number of trials needed to achieve the first success.
- The mean and the variance:

$$E(X) = \frac{1}{p}$$
$$V(X)\ X = \frac{q}{p^2}$$

Negative Binomial Distribution

- This distribution describes the number of trials needed to achieve a fixed number k of successes.

- The probability mass function $p(y)$ for the negative binomial distribution is given by $\binom{y-1}{k-1} q^{y-k} p^{k}$ where y represents the total number of trials needed to achieve k successes.
- The negative binomial distribution can be thought of as the sum of k independent geometric random variables.
- Mean and variance:

$$E(X) = \frac{k}{p}$$

$$V(X) = \frac{kq}{p^2}$$

Example

Question: *Forty percent of the assembled ink-jet printers are rejected at the inspection station. Find the probability that the first acceptable ink-jet printer is the third one inspected. Considering each inspection as a Bernoulli trial with q = 0.4 and p = 0.6 yields*

- To find the probability that the first acceptable ink-jet printer is the third one inspected, we can model each inspection as a Bernoulli trial with a success probability $p = 0.6$ (acceptable printer), and failure probability $q = 0.4$ (rejected printer)
- Using the geometric distribution, we compute $p(3)$, which represents the probability that the first acceptable printer is found on the third inspection. This is calculated as $p(3) = (0.4)^{3-1}(0.6) = 0.096$
- So, in approximately 9.6% of cases, the first acceptable printer is found on the third inspection.
- To find the probability that the third printer inspected is the second acceptable printer, we use the negative binomial distribution. Here, we want to determine $p(3)$, which represents the probability of finding the second acceptable printer on the third inspection. This is calculated as

$$\binom{3-1}{2-1} 0.4^{3-2} 0.6 = 0.288$$